

**CENTRE FOR DEVELOPMENT OF IMAGING TECHNOLOGY
(TECHNOLOGY EXTENSION DIVISION)**

SYLLABUS

CERTIFICATE IN CAD TECHNOLOGY(CCAD)	
Scheme: 2023 Scheme Course Code: 23C03 Duration: 3 Months (250Hrs) Eligibility: SSLC or Equivalent	
THEORY SUBJECTS	
23C03-101	Paper I - Introduction to Information System
23C03-102	Paper II - CAD & Tools
LAB SUBJECTS	
23C03-103	Lab I - CAD & Tools
ANNEXURE	
Assignments for Theory & Questions for Lab	

SYLLABUS FOR THEORY

23C03-101 - INTRODUCTION TO INFORMATION SYSTEM

Theory Hours - 50 Hrs

Module I - Computer Fundamentals (10Hrs)

Learning Outcome

- Understanding of the concept and architecture of computers.
- How a computer system works?
- Learning the basic terminologies.

Topics

Introduction :- Fundamentals - Definition, Hardware, Software, Operating System, data, information etc; Functions and Characteristics; Applications of Computers; Capabilities and Limitations

Familiarise different generations of computer

Computer classifications :- Analogue / Digital / Hybrid, PC, Microcomputer, Minicomputer, Mainframe computer, Super Computer.

Components of a computer :- Input Unit ; Output Unit ; CPU ; Memory unit

Input devices :- Keyboard; Mouse; Scanner; Digital/ Web Camera; Touch screen; graphic tablet

Output Devices :- VDU; Printer; Audio Output devices; LCD Projector

Intelligent terminal and dumb terminal

Memory Unit ; Volatile and Non-Volatile memory ; Primary Memory ; Secondary Memory ; BIOS Memory ; Cache Memory ; Virtual memory ; SIMM ; DIMM ; Storage devices; online / offline storage.

Familiarisation of other hardware components :- SMPS ; Motherboard ; Video Cards - PCI, PCI Express, and AGP ; Multimedia Cards - Sound Cards, TV Tuner Cards and Capture Cards ; I/O Cards - SCSI, Serial, USB, and Parallel ; Communications Cards - NICs and Modems

Ports :- Characteristics and different types - Serial, Parallel, USB, Ethernet port etc

Configuring the Peripheral Devices ; methods for connecting peripherals to system - different types of cable connection, explain Bluetooth and Wi-Fi / IR, NFC connectivity

Data representation in computers :- Number System - Types, examples, Conversion rules ; Coding system - ASCII, BCD, EBCDIC, ISCII, Parity Bit, etc.

Units :- Time Units - Milliseconds, Microseconds, Nanoseconds, Pico seconds ; storage units - Bit, Byte, Kilo Byte, Mega Byte, Giga Byte, Tera Byte, Peta Byte, Exa Byte, Nibble.

Module II – Basics of Information Technology (10Hrs)

Learning Outcome

- Acknowledge the role of technologies in today's society
- To identify the basic hardware and software components of a computer and explore their functioning
- Learn about various operating systems

Topics

Introduction to IT – Role of IT in society - Advantages and Disadvantages, Characteristics and features of IT, Application areas - Science and Engineering, Education, Business and Commerce, Medicine and Entertainment – Advantages & Disadvantages of IT

Basics of Computer System – Generations & Peripheral devices – Hardware – Software – system software & application software-off the shelf and custom software-programming languages-compilers-interpreters- Operating systems: structure & functions – Types of Operating system: Batch processing OS, Mutitasking OS, Multiprocessing OS, Time sharing OS, Real Time OS, Distributed OS, Network OS-Examples of OS: MS-DOS, MS-Windows, Linux Ubuntu, Apple Macintosh

Data and information Concepts -- Importance of data and information, Difference between data and information

Introduction to ITES (IT Enabled Services) – what are IT enabled services? Benefits and threats, ITES fields - BPO – GIS – Call Centers – Data Management Services – Medical Transcription – Data Digitization, Web services, Payroll management, Telemarketing

Module III – Basics of Software (10Hrs)

Learning Outcome

- Distinguish System software and Application software.
- Identify the primary functions of Operating Systems.
- Installation and configuration of software.

Topics

Types of Software :- Software - System software , Application software, Utility Software
Demo ware ; Shareware ; Freeware ; Firmware ; Free and Open Source Software.

Operating Systems :- Functions, Types - Batch Processing , Single User, Multi User,
Multiprogramming, Multi-Tasking (Multiprocessing); Familiarise Multithreading
Booting and Shutdown of Computers ; BIOS ; POST ; Master Boot Record(MBR)

Introduction to old versions of Windows OS

Detailed study of Windows :- Overview ; GUI basics (Desktop, taskbar, icons, desktop
background management) ; Navigation (Start menu, File explorer) ; Accessories ; Control Panel
; User Account management ; storage management ; file and folder management ; keyboard
shortcuts ; data backup

Familiarise Ubuntu Linux :- Introduction ; History ; Versions ; basics

Familiarising Command mode :- Windows (DOS) ; Linux (Terminal)

Familiarize other operating systems :- UNIX ; Apple Macintosh

Linux commands

Computer Virus :- Introduction ; Mechanism of Virus ; How a virus spreads ; How is virus
named ; A few prominent viruses ; Types of Computer Virus ; Related Concepts - Antivirus
programs - Norton antivirus program

Module IV - Basics of Information System (10Hrs)

Learning Outcome

- Explain how is an information system is helpful for an organization
- Explain the role of business information systems
- Write the advantages of MIS
- Explain an Enterprise System

Topics

Information System :- Data, Information & System ; Information System; Dimensions of Information System - Organization, Management, IT ; Functions of Information System

Business Information Systems :- Role of Business Information Systems - Top, Middle, Bottom ;

Types of Business Information Systems - TPS, MIS, EIS, DSS, AI/Expert Systems

Management Information System(MIS) :- Objectives ; Components ; Functions ; Resources ; Characteristics; Terminologies and Applications

Enterprise System and Types :- ERP ; CRM ; SCM

Module V – Social Media & Cyber Security (10Hrs)

Learning Outcome

- Understand various social media technologies and their importance in the day to day life.
- Benefits of social media in digital marketing and Education,
- Awareness on various cybercrimes and cyber security,
- Understand copyright and intellectual property rights

Topics

Social Media – Why & what Social Media – Present and Past - Social media Platforms and applications – Usage in business & Education- social media etiquette

Social media and digital marketing

Information Security – Concepts of Security - two factor authentication(2FA) - Biometric authentication methods - Data Encryption and Decryption- Cryptography

Awareness about cyber crime and Cyber security

Cyber crime - What is cyber crime, Understand the term computer Virus and Malware, Email spamming, Email spoofing, Email bombing, Forgery, Hacking, Cyber Stalking or Bullying, Web jacking, Phishing, Pornography, Cookies and threats

Cyber security - Basic concepts - Elements of security - Data security and protection from cyber attack - Antivirus - password, Concept of Firewall, Digital signature - uses and working

Cyber Laws - An Overview & Scope of Cyber Law - Copyright, Copyleft ,Trademarks and Software Patenting - Data Privacy & Confidentiality - Cyber laws in India - IT ACT-objectives, features, offences

23C03-102 – CAD & Tools

Theory Hours - 50 Hrs

Module I – AutoCAD - Introduction (10Hrs)

Learning Outcome

- Understand the purpose and scope of AutoCAD in engineering and design industries with key features
- Apply basic principles of graphics, geometry and mathematical operations to create and manipulate objects in AutoCAD.
- Familiarize AutoCAD Interface and Drawing Tools, Navigation, Control, and Inquiry

Topics

AUTOCAD - Introduction and Background - Pre-Flight Checks - Brief History - Graphics Primer - Math Background-The Basics - Scale - The AUTOCAD Interface - Drawing Objects - Lines - Circles -Text - Arcs - Rectangles - Complex Entities - Navigation & Control - World Coordinate System - Zoom & Pan - Snapping - Layers - Colors - Linetypes - List & inquiry

Module II – AutoCAD - Editing (10 Hrs)

Learning Outcome

- Understand the basic principles and importance of editing within AutoCAD.
- Familiarize with Object Selection and Manipulation, Advanced Object Editing Techniques
- Understand the Workflow Optimization and Output generation

Topics

Editing - Selecting Objects - Copying Objects -Moving Objects - Resizing Objects - Editing Objects - Trim and Extent – Grips- **Power Tools** - Blocks -Hatch Patterns - Arrays -Mirror - Filtering Selections – Groups - **Printing & Plotting** - Model Space - Paper Space

Module III – AutoCAD – 2D Drawing – Part 1 (10 Hrs)

Learning Outcome

- Understand the principles and applications of 2D drawing in AutoCAD.
- Familiarize with Formatting and Standards in AutoCAD, Visual Properties and Styles
- Understand File Management and Collaboration in AutoCAD

Topics

Drawing - Straight lines - Rectangles- Circles - Arcs - Polygons - Ellipse - Text Creation - Hatching, etc-Formatting - Units Formats - Dimensions styles - creating Layers - line type - Object's Colors - Multiline Styles Line - weight settings - Point Styles ,etc
Files - Opening and Saving Files - Exporting Files - Drawing recovery - auditing and recovering the error files - purging the unwanted blocks - layers - dimension styles, etc

Module IV - AutoCAD - 2D Drawing - Part 2 (10Hrs)

Learning Outcome

- Understand the principles of automatic dimensioning in AutoCAD.
- Familiarize with Editing Commands and Modifications and Refinements features
- Utilize AutoCAD tools for spell checking and performing inquiries.

Topics

Automatic Dimensioning - Vertical / Horizontal Dimensions - Inclined Dimensions - Radius of circles - Diameter of Circles - Tolerance, text and lines alignment
Editing - Copy - paste - paste special - copy with base point -paste to original coordinate , etc
Modifications - Erase - Copy Objects - Move objects from one place to another - Text edit - extend lines and arcs to a boundary- trim objects within a boundary - change object's within a boundary - change object's property [colour , line type, thickness, etc]- edit hatches, edit texts
Tools- Spell checking - enquiries like area - distance and volumes - table settings,etc

Module V - AutoCAD - 3D Drawing (10 Hrs)

Learning Outcome

- Understand the principles and techniques of 3D drawing in AutoCAD.
- Configure AutoCAD options to personalize the user interface
- Familiarization with Customization of Menus and Toolbars

Topics

Drawing - Drawing solids like Boxes, wedges, sphere, cylinder, cone, Torus, and 3D surfaces,etc
Modifications - solid editing like 3D Rotating,3D Array,merging of more than two objects
AUTOCAD - OPTIONS and CUSTOMISATION
Tools-Options like changing AUTOCAD background colour, fonts of menus,setting of automatic saving time,saving open files locations,changing or setting the template file locations,setting the temporary file locations,set the saving file type,set the plotter,set the default line weight,etc ,Custmisation of menus,tool bars,etc

DETAILED SYLLABUS (LAB)

SYLLABUS FOR LAB

23C03-103 LAB I - CAD & Tools

Lab Hours - 150 Hrs

Module I

1. Familiarize with the Installation and Pre-Flight Checks for AutoCAD
2. Understanding 2D and 3D coordinate systems, Cartesian coordinates, basic Mathematics for AutoCAD
3. Work with AutoCAD interfaces and basic drawing tools, text and animation, complex entities
4. Understanding the WCS and using coordinate entry methods.
5. Practice with Precision Tools and Drawing Management
6. Understanding visual enhancement for clarity and readability

Module II

1. Understand the basics of AutoCAD editing and basic editing commands
2. Familiarize with Object Selection and Manipulation, Advanced Editing Techniques
3. Work with features like Resizing and Rotating Objects, Mirror and Offset Commands, Groups and Blocks
4. Familiarize the methods for printing the edited

Module III

1. Practice creating and editing basic shapes to understand fundamental drawing tools.
2. Understanding unit formats and their impact on drawing accuracy.
3. Familiarize with Text Creation and Annotation, Hatching and Fills, Visual Properties and Styles, Multiline Styles and Point Styles
4. Exporting and Importing drawings to various file formats for sharing and collaboration.

Module IV

1. Practice creating basic shapes and exploring drawing precision using coordinate input.
2. Understanding automatic dimensioning tools in AutoCAD.
3. Utilizing essential editing commands for exploring and modifying text and entities.
4. Utilize AutoCAD tools for spell checking and performing inquiries.

Module V

1. Introduction to 3D coordinate systems and navigation tools.
2. Familiarizing Creating complex 3D solids and Solid Modeling and Editing
3. Understanding Customization of AutoCAD Options, Customization of Menus and Toolbars, Advanced Tools and Techniques

ASSIGNMENTS FOR THEORY AND QUESTIONS FOR LAB

ASSIGNMENTS FOR THEORY

Paper 1(Theory)

23C03-101- INTRODUCTION TO INFORMATION SYSTEM

Module I – Computer Fundamentals

1. History and Generations of Computers.
2. Explain in detail about various Input Output devices
3. Explain about different types of storage devices with capacity (based on purpose)
4. Specifications for new personal computer purchase
5. Show the computations needed for the conversion between various number systems

Module II – Basics of Information Technology

1. Explain how information systems are related with modern life
2. Role of hardware and software in computing
3. Explain the role of information systems in education
4. Discovering the milestones of Information & Communication Technology history

Module III – Basics of Software

1. Discuss the pros and cons of the three major operating systems.
2. What are the different versions of Windows OS
3. Updating operating system and web browser software
4. Explain about various DOS and Linux commands
5. Various methods for backing up of user data and disc management
6. Computer Virus and Green IT

Module IV - Basics of Information System

1. Explain how is an information system is helpful for an organization
2. Explain the role of business information systems
3. Write the advantages of MIS
4. Explain an Enterprise System

Module V – Social Media & Cyber Security

1. Explain the elements of cyber security
2. Explain the impact of social media in modern life
3. Explain the working and the advantages of Digital Signature
4. How to market a product using social media?
5. How to provide privacy and security features in your different social media accounts
6. Explain amendments in IT act.
7. What do you mean by fact checking?
8. Explain about http and https.
9. Explain about URL and SSL certification

Paper 2(Theory)

23C03-102 – CAD & TOOLS

Module I – AutoCAD- Introduction

1. Describe the system requirements and typical pre-flight checks that must be met before installing AutoCAD. Why do these checks matter?
2. Describe the significance of scale in AutoCAD drawings. Give examples of design projects when scale is essential.
3. Examine and contrast the ways in which various AutoCAD object types can be created and edited.
4. Discuss about how snapping tools are essential for precise object alignment and connectedness when drawing.
5. Describe the importance of line types and colours in AutoCAD designs. In what ways do these characteristics improve readability and comprehension of designs?

Module II – AutoCAD- Editing

1. Explain the significance of AutoCAD's editing features. Describe the ways in which efficient and productive editing methods improve design operations.
2. Discuss about the importance of AutoCAD's object selection techniques. Describe the distinctions between applying filters based on object properties, employing selection sets, and choosing objects one at a time.
3. Examine the differences between AutoCAD's Model Space and Paper Space. Describe the goals of each and the process by which drawings are ready for printing and plotting in each area.
4. Describe the functions and uses of AutoCAD's advanced editing tools, such as the Trim and Extend commands. Show how to extend and change lines and objects in a drawing using these instructions.
5. Explain what AutoCAD Blocks mean. Discuss how Blocks are created, edited, and reused to streamline design processes and maintain consistency across drawings

Module III – AutoCAD 2D Drawing – Part 1

1. Explain the purpose of an AutoCAD 2D sketch. How can fundamental geometric shapes like lines, rectangles, circles, arcs, polygons, and ellipses be created more easily with AutoCAD?

2. Discuss about how text generation in AutoCAD drawings is important. How could labels and annotations improve the readability and comprehension of technical drawings?
3. Describe the function of AutoCAD's unit formats. How can unit settings and adjustments affect the precision and coherence of drawings?
4. What factors need to be taken into account while selecting file formats for cooperation and compatibility?
5. Discuss the significance of AutoCAD cleansing. How can drawing performance be enhanced and file size optimised by removing unnecessary layers, blocks, and dimension styles? Describe why AutoCAD objects have different colours and types of lines assigned to them.

Module IV - AutoCAD 2D Drawing - Part 2

1. In AutoCAD, compare and contrast inclined, vertical, and horizontal dimensions.
2. Discuss the significance of Copy with Base Point and Paste to Original Coordinates commands in precise object placement and duplication tasks.
3. Describe the function of AutoCAD's spell-checking features. What role does spell checking have in preserving the professionalism and accuracy of drawings?
4. Discuss the importance of area, distance, and volume inquiries in AutoCAD. How are these tools useful in design verification and analysis?
5. Explain how you would use AutoCAD to extend lines and arcs to a boundary. What factors need to be taken into account while doing this?
6. Describe the procedures for utilising AutoCAD's Trim command to trim items inside a border. How does this command help in refining drawing details

Module V - AutoCAD 3D Drawing

1. Explain how to use AutoCAD to create 3D solids.
2. Describe the concept of AutoCAD solid editing. Talk about the use of three solid editing commands to alter 3D models.
3. Explain on the importance of inputting default line weights and template file locations in AutoCAD. In what ways may these configurations enhance project consistency and standardise design practices?
4. Explain how to personalise AutoCAD toolbars and menus. Why is customisation crucial for CAD software workflow and efficiency optimisation?
5. Describe the function and use of AutoCAD's advanced capabilities, such as plotter settings, area inquiries, and spell checking. What role do these tools play in project management and accurate design?

QUESTIONS FOR LAB

23C03-103 LAB I – CAD & TOOLS

Hours - 150 Hrs

Module I

1. Create a 2D drawing using AutoCAD, specifying precise coordinates for points using Cartesian coordinates and polar coordinates. Apply scaling principles to resize the drawing without altering its proportions. Measure and verify dimensions before and after scaling.
2. Draw a floor plan of a simple room using AutoCAD. Include walls, windows, and doors with geometrical shapes. Label each element with appropriate dimensions.
3. Design a company logo. Apply text elements into the logo using different text styles and orientations.
4. Draw a 2D landscape with various elements using the WCS. Demonstrate how to move and rotate objects relative to the WCS.

Module II

1. Use basic editing commands to modify a simple room plan created earlier. Ensure all dimensions and object placements are accurate.
2. Create a selection set using various methods to highlight specific elements in a building layout.
3. Replicate furniture objects in a room layout. Adjust the placement of copied objects to maintain consistent spacing and alignment.
4. Resize and rotate doors and windows in a building facade to fit specific architectural dimensions and design requirements.
5. Mirror a complex architectural element across a central axis. Use the Offset command to create parallel copies of supporting columns. Create a group of furniture elements in a meeting room layout. Convert the group into a block for easy replication across different rooms.
6. Set up a layout in Paper Space & Model Space for printing room plan. Include title blocks, scales, and necessary annotations for presentation.

Module III

1. Create a floor plan of a room using straight lines for walls, rectangles for doors and windows, and circles for light fixtures.
2. Label different areas of the floor plan created in above practise question using text creation tools. Include room names, dimensions, and other annotations.

3. Create multiple layers for different elements of the floor plan. Assign colors and linetypes to each layer.
4. Create a complex multiline style to represent a detailed electrical diagram or plumbing layout within the floor plan.

Module IV

1. Create a floor plan of a room including walls, doors, windows, and furniture using basic drawing tools. Apply automatic dimensioning for all elements.
2. Design a mechanical part drawing with inclined surfaces and intricate geometric shapes. Apply radius and diameter dimensions to circles and inclined dimensions to angular features.
3. Modify the floor plan created in Exercise 1 by copying and pasting furniture elements to different locations. Use the Move command to relocate doors and windows. Edit text annotations to update room labels.
4. Perform a spell check on all text annotations within the drawing. Use AutoCAD inquiries to calculate the area of specific rooms and the distance between key elements.

Module V

1. Using the 3D drawing tools in AutoCAD, create a 3D model that has a box, wedge, sphere, and cylinder.
2. Design a composite 3D solid model that combines a cone, torus, and pyramid. Apply different solid modeling techniques such as extrusion and revolution.
3. Modify the 3D model created in Exercise 1 by rotating, scaling, and mirroring selected objects. Use the 3D Array command to create a pattern of objects.
4. Customize AutoCAD options such as changing background colors, interface themes, and setting automatic saving intervals and file locations.
5. Create custom menus, toolbars, or ribbon panels in AutoCAD to organize frequently used commands and tools for 3D modeling.
6. Perform a spell check on text annotations within the 3D model. Use area inquiries to calculate volumes and surface areas of selected objects.